CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 4 Solutions

DP, greedy and B-trees · Lectures 4 to 7

Muharib

01 Solutions Q1 and Q2

#	Answer	Reason
1	**True**	
2	**False**	`O(2ⁿ)`; the DP version is `O(n)`.
3	**False**	Greedy never reverses a decision.
4	**False**	The slide example: 160 instead of 220.
5	**True**	
6	**True**	
7	**True**	
8	**False**	At most `2t − 1` keys.
9	**True**	
10	**False**	For disks (secondary storage).

Q2: 1. c (10·20·5) 2. a 3. b 4. b 5. b 6. c

02 Solution Q3: matrix chain

```
p = 30, 35, 15, 5, 10

AB = 30·35·15 = 15,750     BC = 35·15·5 = 2,625     CD = 15·5·10 = 750

A(BC) = 2,625 + 30·35·5 = 7,875   ✓

(AB)C = 15,750 + 30·15·5 = 18,000

B(CD) = 750 + 35·15·10 = 6,000

(BC)D = 2,625 + 35·5·10 = 4,375   ✓

ABCD: A(BCD)   = 4,375 + 30·35·10 = 14,875
      (AB)(CD) = 15,750 + 750 + 30·15·10 = 21,000
      (ABC)D   = 7,875 + 30·5·10 = 9,375   ✓
```

	A	B	C	D
A	0	15750	7875	**9375**
B	–	0	2625	4375
C	–	–	0	750
D	–	–	–	0

```
Minimum = 9,375        Order: ((A(BC))D)
```

03 Solution Q4: LCS

	∅	B	D	C	B
∅	0	0	0	0	0
B	0	**1**	1	1	1
A	0	1	1	1	1
C	0	1	1	**2**	2
D	0	1	2	2	2
B	0	1	2	2	**3**

Length = 3 LCS = BCB (BDB is also correct)

04 Solutions Q5 and Q6

Q5

sorted by f already

take a1 (f = 3)

a2 (s = 2 < 3) ✗ a3 (s = 4 ≥ 3) ✓ take, f = 7

a4 (s = 1) ✗ a5 (s = 5 < 7) ✗

a6 (s = 8 ≥ 7) ✓ take, f = 10

a7 (s = 9 < 10) ✗

Answer: {a1, a3, a6} → 3 activities

Q6

Item	w	v	$\frac{v}{w}$
2	4	40	10
1	5	30	6
3	6	30	5
4	3	6	2

a) Fractional: item 2 (4 kg, 40) + item 1 (5 kg, 30)

+ $\frac{1}{6}$ of item 3 (1 kg, 5)

total = 40 + 30 + 5 = 75

b) 0-1 greedy: item 2 + item 1 = 70 (1 kg left; nothing else fits)

Best 0-1: item 2 + item 3 (10 kg) → 70

item 1 + item 2 (9 kg) → 70

So greedy gives 70 here, which happens to be optimal.

05 Solution Q7: Huffman

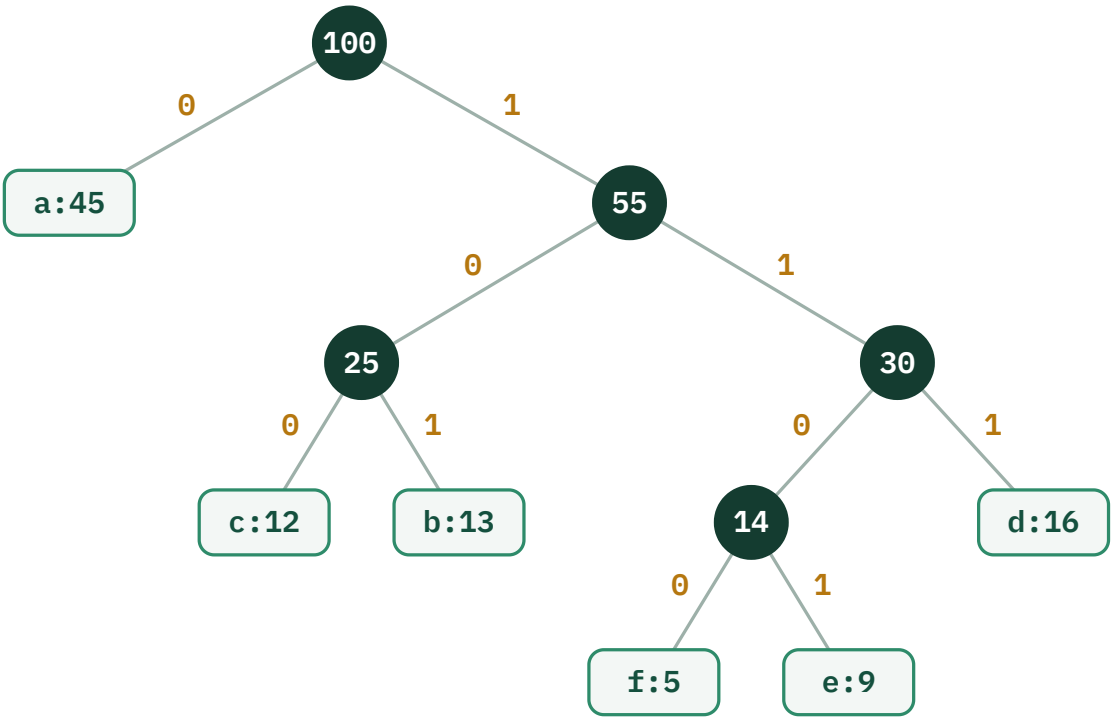
f:5 + e:9 = 14

c:12 + b:13 = 25

14 + d:16 = 30

25 + 30 = 55

a:45 + 55 = 100 (root)



Letter	Freq	Code	Bits
a	45	0	45
b	13	101	39
c	12	100	36
d	16	111	48
e	9	1101	36
f	5	1100	20

```
Huffman: 45 + 39 + 36 + 48 + 36 + 20 = 224 bits
```

```
Fixed 3-bit code: 100 × 3 = 300 bits
```

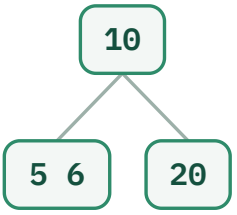
06 Solution Q8: B-tree insert (t = 2)

t = 2: at most 3 keys. A node with 3 keys is split on the way down.

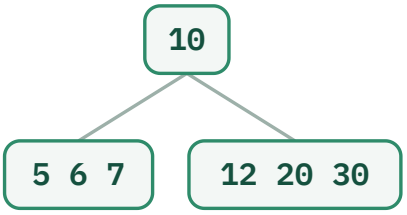
10, 20, 5 → one full node:



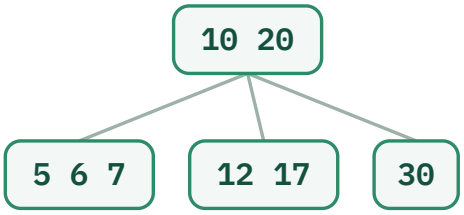
6: the root is full → split around 10, then 6 goes left:



12, 30, 7 without a split:



17: child `12 20 30` is full → split, 20 moves up, and 17 goes into `12 17`:



07 Solution Q9: delete C

C is in internal node `C G M`. The child before it, `A B`, has 2 = t − 1, and the child after it,

`D E F`, has 3 ≥ t → **Case 2b**: replace C with its successor D and delete D from the leaf.

